

**RAJALAKSHMI ENGINEERING COLLEGE**

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AD23431 - STATISTICAL ANALYSIS AND COMPUTING

EXP 3: Implementation of List, Factors, and Data Frames in R

1. Employee Information Management Using Lists

An organization stores heterogeneous employee details such as **ID, Name, Salary, Department, and Joining Date**.

Using Lists in R, do the following:

- Create a list to store employee information.

```
employee <- list(  
  
  ID = 101,  
  
  Name = "Rohan",  
  
  Salary = 35000,  
  
  Department = "AIDS",  
  
  Joining_Date = "2023-06-15"  
)
```

```
employee
```

OUTPUT:

```
employee
```

```
$ID
```

```
[1] 101
```

```
$Name
```

```
[1] "Rohan"
```

```
$Salary
```



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[1] 35000

\$Department

[1] "AIDS"

\$Joining_Date

[1] "2023-06-15"

- b. Access and modify the employee salary.

employee\$Salary

employee\$Salary <- 40000

employee

OUTPUT:

employee

\$ID

[1] 101

\$Name

[1] "Rohan"

\$Salary

[1] 40000

\$Department

[1] "AIDS"

\$Joining_Date

[1] "2023-06-15"

- c. Add a new element for **Performance Rating**.



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```
employee$Performance_Rating <- "Excellent"
```

```
employee
```

OUTPUT:

```
employee
```

```
$ID
```

```
[1] 101
```

```
$Name
```

```
[1] "Rohan"
```

```
$Salary
```

```
[1] 40000
```

```
$Department
```

```
[1] "AIDS"
```

```
$Joining_Date
```

```
[1] "2023-06-15"
```

```
$Performance_Rating
```

```
[1] "Excellent"
```

- d. Remove an element from the list.

```
employee$Department <- NULL
```

```
employee
```

OUTPUT:

```
employee
```

```
$ID
```

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*[1] 101**\$Name**[1] "Rohan"**\$Salary**[1] 40000**\$Joining_Date**[1] "2023-06-15"**\$Performance_Rating**[1] "Excellent"*

- e. Convert the list into a data frame.

*employee_df <- as.data.frame(employee)**employee_df***OUTPUT:***employee_df**ID Name Salary Joining_Date Performance_Rating**101 Rohan 40000 2023-06-15 Excellent*

2. Student Academic Record Using Nested Lists

A student's academic details include **personal information** and **semester-wise marks**.

Using Lists in R, do the following:

- a. Create a nested list containing student details and semester-wise marks.

student <- list(

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```
Personal_Info = list(  
  
Roll_No = 231,  
  
Name = "Rohan",  
  
Department = "AI & DS"  
  
),  
  
Marks = list(  
  
Semester1 = c(Maths = 85, Physics = 78, Programming = 88),  
  
Semester2 = c(Maths = 90, Statistics = 82, Data_Structures = 86)  
  
)  
  
)  
  
student  
  
OUTPUT:  
  
student  
  
$Personal_Info  
  
$Personal_Info$Roll_No  
  
[1] 231  
  
$Personal_Info$Name  
  
[1] "Rohan"  
  
$Personal_Info$Department  
  
[1] "AI & DS"  
  
$Marks  
  
$Marks$Semester1
```

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Maths Physics Programming**85 78 88*****student\$Marks\$Semester2*****Maths Statistics Data_Structures****90 82 86**

- b. Extract marks of a specific semester.

student\$Marks\$Semester1***OUTPUT:*****Maths Physics Programming****85 78 88**

- c. Calculate the average marks across all semesters.

all_marks <- c(student\$Marks\$Semester1, student\$Marks\$Semester2)***average_marks <- mean(all_marks)******average_marks******OUTPUT:******average_marks******[1] 84.83333***

- d. Update the marks of one subject.

student\$Marks\$Semester2["Maths"] <- 95***student\$Marks\$Semester2******OUTPUT:******student\$Marks\$Semester2***


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Maths Statistics Data_Structures

95 82 86

e. Display the structure of the list.

str(student)

OUTPUT:

str(student)

List of 2

\$ Personal_Info:List of 3

..\$ Roll_No : num 231

..\$ Name : chr "Rohan"

..\$ Department: chr "AI & DS"

\$ Marks :List of 2

..\$ Semester1: Named num [1:3] 85 78 88

.. ..- attr(, "names")= chr [1:3] "Maths" "Physics" "Programming"*

..\$ Semester2: Named num [1:3] 95 82 86

.. ..- attr(, "names")= chr [1:3] "Maths" "Statistics" "Data_Structures"*

3. Customer Feedback Analysis Using Factors

A café collects customer feedback categorized as **Excellent**, **Good**, **Average**, and **Poor**.

Using Factors in R, do the following:

a. Create a factor variable for customer feedback.

feedback <- factor(c("Excellent", "Good", "Average", "Poor",


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"Good", "Excellent", "Average", "Good")

feedback

OUTPUT:

feedback

[1] Excellent Good Average Poor Good Excellent Average

[8] Good

Levels: Average Excellent Good Poor

- b. Display factor levels and their frequency.

levels(feedback)

table(feedback)

OUTPUT:

levels(feedback)

[1] "Average" "Excellent" "Good" "Poor"

table(feedback)

feedback

Average Excellent Good Poor

2 2 3 1

- c. Reorder the factor levels based on sentiment.

feedback_ordered <- factor(

feedback,

levels = c("Poor", "Average", "Good", "Excellent"),

ordered = TRUE

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)

*feedback_ordered***OUTPUT:***feedback_ordered**[1] Excellent Good Average Poor Good Excellent Average**[8] Good**Levels: Poor < Average < Good < Excellent*

d. Convert the factor into a numeric scale for analysis.

*feedback_numeric <- as.numeric(feedback_ordered)**feedback_numeric***OUTPUT:***feedback_numeric**[1] 4 3 2 1 3 4 2 3*

e. Check whether the factor is ordered or unordered.

*is.ordered(feedback)**is.ordered(feedback_ordered)***OUTPUT:***is.ordered(feedback)**[1] FALSE**is.ordered(feedback_ordered)**[1] TRUE*

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4. Employee Shift Scheduling Using Factors

Employees work in **Morning, Evening, and Night** shifts.

Using Factors in R, do the following:

- a. Create an ordered factor for work shifts.

```
shifts <- factor(  
  c("Morning", "Evening", "Night", "Morning",  
    "Night", "Evening", "Morning"),  
  levels = c("Morning", "Evening", "Night"),  
  ordered = TRUE  
)
```

shifts

OUTPUT:

shifts

[1] Morning Evening Night Morning Night Evening Morning

Levels: Morning < Evening < Night

- b. Change the reference level of the factor.

```
shifts_ref <- factor(  
  shifts,  
  levels = c("Evening", "Morning", "Night"),  
  ordered = TRUE  
)
```

shifts_ref

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OUTPUT:*shifts_ref**[1] Morning Evening Night Morning Night Evening Morning**Levels: Evening < Morning < Night*

- c. Count the number of employees per shift.

*table(shifts)***OUTPUT:***table(shifts)**shifts**Morning Evening Night**3 2 2*

- d. Convert the factor to character type.

*shifts_char <- as.character(shifts)**shifts_char***OUTPUT:***shifts_char**[1] "Morning" "Evening" "Night" "Morning" "Night" "Evening" "Morning"*

- e. Explain the importance of ordered factors in analysis.

*>>Ordered factors show ranking or order**>>Help in comparison between levels**>>Useful for statistical analysis**>>Improve interpretation of results*



5. Student Performance Analysis Using Data Frames

A class maintains records of **Roll No, Name, Marks in Three Subjects, and Result**.

Using Data Frames in R, do the following:

- a. Create a data frame for student performance.

```
student <- data.frame(  
  RollNo = c(1, 2, 3, 4),  
  Name = c("Praji", "Fayaz", "Pravesh", "Rohan"),  
  Subject1 = c(78, 65, 82, 90),  
  Subject2 = c(75, 72, 88, 85),  
  Subject3 = c(80, 68, 79, 92),  
  Result = c("Pass", "Pass", "Pass", "Pass")  
)  
student
```

OUTPUT:

student

<i>RollNo</i>	<i>Name</i>	<i>Subject1</i>	<i>Subject2</i>	<i>Subject3</i>	<i>Result</i>
1	Praji	78	75	80	Pass
2	Fayaz	65	72	68	Pass
3	Pravesh	82	88	79	Pass
4	Rohan	90	85	92	Pass

- b. Add a new column for **Total Marks**.

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```
student$Total <- student$Subject1 +
```

```
    student$Subject2 +
```

```
    student$Subject3
```

```
student$Total
```

OUTPUT:

```
student$Total
```

```
[1] 233 205 249 267
```

- c. Filter students who scored more than **70** in all subjects.

```
high_scorers <- student[
```

```
  student$Subject1 > 70 &
```

```
  student$Subject2 > 70 &
```

```
  student$Subject3 > 70, ]
```

```
high_scorers
```

OUTPUT:

```
high_scorers
```

```
RollNo  Name Subject1 Subject2 Subject3 Result Total
```

```
1 Praji    78    75    80 Pass  233
```

```
3 Pravesh  82    88    79 Pass  249
```

```
4 Rohan   90    85    92 Pass  267
```

- d. Sort students based on total marks.

```
student_sorted <- student[order(student$Total, decreasing = TRUE), ]
```

```
student_sorted
```

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OUTPUT:*student_sorted***RollNo Name Subject1 Subject2 Subject3 Result Total****4 Rohan 90 85 92 Pass 267****3 Pravesh 82 88 79 Pass 249****1 Praji 78 75 80 Pass 233****2 Fayaz 65 72 68 Pass 205**

- e. Export the data frame to a CSV file.

write.csv(student, "student_performance.csv", row.names = FALSE)

6. Sales Dataset Exploration Using Data Frames

A retail store maintains daily sales data including **Date, Product, Quantity, Price, and Total Amount**.

Using Data Frames in R, do the following:

- a. Create a data frame for sales data.

sales <- data.frame(***Date = c("2024-01-01", "2024-01-01", "2024-01-02", "2024-01-02"),******Product = c("Pen", "Book", "Pen", "Pencil"),******Quantity = c(10, 5, 8, NA),******Price = c(10, 50, 10, 5)******)******sales\$TotalAmount <- sales\$Quantity * sales\$Price***

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*sales***OUTPUT:***sales***Date Product Quantity Price TotalAmount****1 2024-01-01 Pen 10 10 100****2 2024-01-01 Book 5 50 250****3 2024-01-02 Pen 8 10 80****4 2024-01-02 Pencil NA 5 NA**

- b. Compute total sales amount per product.

aggregate(TotalAmount ~ Product, data = sales, sum, na.rm = TRUE)**OUTPUT:****Product TotalAmount****Book 250****Pen 180**

- c. Identify the best-selling product.

best_product <- aggregate(TotalAmount ~ Product, data = sales, sum, na.rm = TRUE)***best_product[which.max(best_product\$TotalAmount),]*****OUTPUT:****Product TotalAmount****Book 250**

- d. Handle missing values in the dataset.



```
sales$Quantity[is.na(sales$Quantity)] <- mean(sales$Quantity, na.rm = TRUE)
```

```
sales
```

OUTPUT:

```
sales
```

```
      Date Product Quantity Price TotalAmount
```

```
1 2024-01-01   Pen 10.000000   10      100
```

```
2 2024-01-01  Book  5.000000   50      250
```

```
3 2024-01-02   Pen  8.000000   10       80
```

```
4 2024-01-02 Pencil 7.666667    5       NA
```

- e. Display summary statistics of the data frame.

```
summary(sales)
```

OUTPUT:

```
summary(sales)
```

```
      Date      Product      Quantity      Price
```

```
Length:4      Length:4      Min. : 5.000  Min. : 5.00
```

```
Class :character Class :character 1st Qu.: 7.000 1st Qu.: 8.75
```

```
Mode :character Mode :character Median : 7.833 Median :10.00
```

```
Mean : 7.667 Mean :18.75
```

```
3rd Qu.: 8.500 3rd Qu.:20.00
```

```
Max. :10.000 Max. :50.00
```

```
TotalAmount
```



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Min. : 80.0

1st Qu.: 90.0

Median :100.0

Mean :143.3

3rd Qu.:175.0

Max. :250.0

NA's :1

Learning Outcomes:

Evaluation Procedure	Marks Awarded
Program(5)	
Execution(5)	
Viva(5)	
Total(15)	
Faculty Signature	

Name: Sai Rajaram J

Roll No: 241801238



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